Unit 2 - Computer Systems
Assignment 4 - Operating Systems

Introduction

You work in a large computer store as a Technical Sales person and you advise customers and other members of staff on computer systems. A major part of your job is to give expert advice to customers as to which system to purchase based upon their specific needs. Another part of your job is to train staff members so that they can also advise customers as to the system specification to choose based upon their particular needs

Task 3 (2P3): Utility Software

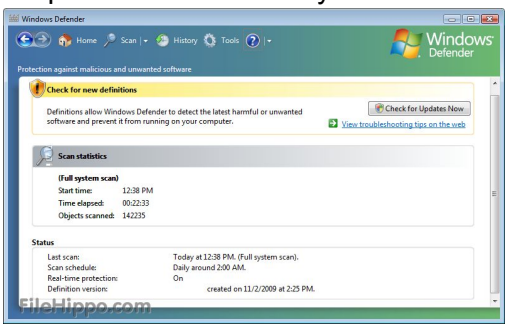
There are a number of software utilities that help the technician to maintain their systems complete the following table to explain the software utilities with a detailed example of its purpose.

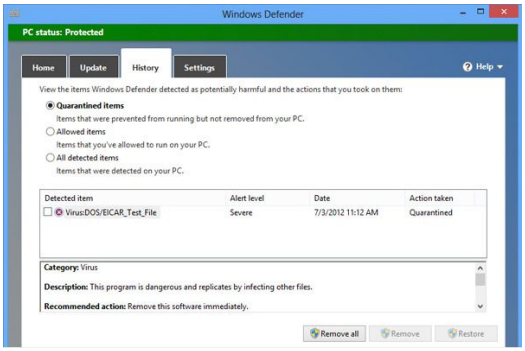
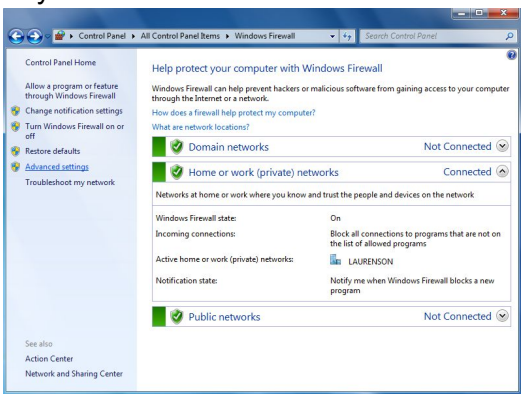
	Explain the Purpose of Utility	**Name of Windows Utility**	**Name of 3rd Party Utility**
Anti-virus Software	The Anti-Virus software is software that you can install on your computer so that you can be alerted when you have a virus. Anti-Virus software will help to keep the PC safe against worms, viruses, Trojan horses and other unwanted programs.	Windows Defender	Norton™ Antivirus
Firewalls	Firewalls, if properly used, can keep your computer safe from harm by setting up filters to detect and block intrusion attempts from external users such as hackers from the internet or any other network.	Windows Firewall	Norton Internet Security™
Clean up tools	Clean up tools helps free up space on the hard disk on the PC by searching the disk for files that you can safely delete. For example, you can choose to delete some or all of the files.	Disk Cleanup	CCleaner
Disk Defragmentation	Disk defragmentation is the process of putting file pieces back together. What disk defragmenters do is rewrite files into contiguous blocks of free space making sure that all file fragments are written in a consecutive order.	Disk Defragmenter	Auslogics Disk Defrag Professional

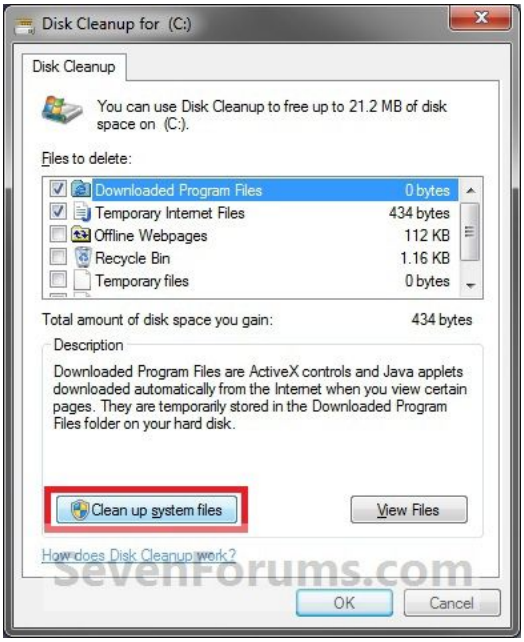
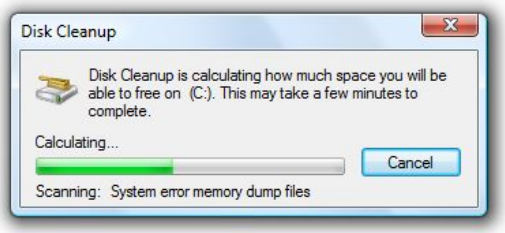
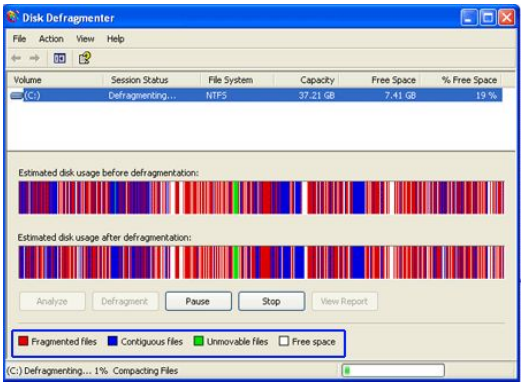
Disk Management	Disk management is used to manage the drives installed in a computer. For example, hard disk drives, optical disk drives and flash drives. Disk management can also be used to partition drives, format drives and assign drive letters.	Disk Management	Paragon Partition Manager™

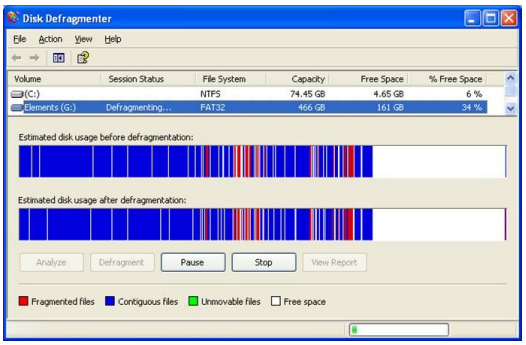
Task 4 (2D1):Use of Utility Software

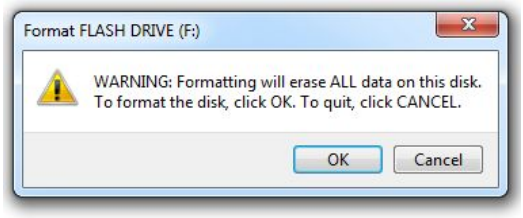
With reference to the utilities (Defragmentation, Disk Cleanup, etc.) identified in task 3 give 2 specific examples of how each utility can be used to improve to the performance of a system. Your detailed example should include reference to improvements in specific performance criteria (e.g. Access time, etc.)

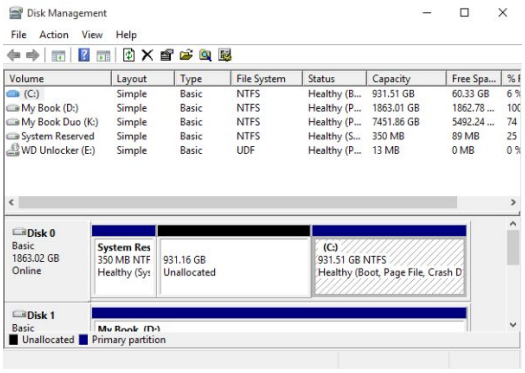
	Name of Windows/3rd Party Utility	**Explanation of how these utilities can improve the performance of a system (Illustrated with annotated screen shots)**
Anti-virus Software	Windows Defender The Image below is a screenshot of Windows Defender with the scan window open to search for malware or other harmful software which impact the performance of the system.  The image below is another anti virus software which has a table showing when the last scans were, this is important for system performance because this action has to be carried out regularly in order to keep system performance at its peak.	The reason that Virus protection improves the performance of a system, is that it maintain a safe environment for the user to use and it keeps out malware that will have the potential to slow down and decrease system performance and effectiveness. It does this by preventing malware from entering a system, detecting any malware that is on a system and ultimately removing that malware from the system. The second reason that the anti virus improves system performance is that it can free the system out of previously attained malware. If the system is malware free then the system can run effectively and without interference or performance decrease, this is because without virus protection malware would go relatively unnoticed on a system unless the user found and deleted the malware, but virus protection does this for the user and also speeds up

		the process as it is designed to perform this action.
Firewalls	Windows Firewall Below is a screenshot of the windows firewall showing that it is turned on and fully functional.  Below is an image of a pop up box of the firewall protecting the system from an unknown external program. 	The first reason that a firewall improves system performance is that this utility helps keep out unwanted traffic that can have the potential to harm the system with malware or malicious code, this helps to maintain a stable system performance by terminating any suspicious data packets that could contain malware or code that could damage the system. The second reason that firewalls improve system performance is that firewalls also stop the external programs, websites or users to steal valuable system information and files which would impact the performance or even corrupt the system itself, causing the system to shut down if such file like the “system32” file was removed.
Clean up tools	Disk Cleanup The image below is a screenshot of the Disk Cleanup menu with the button to start the cleaning process.	The first reason clean up tools improves system performance is that this utility deletes unnecessary files and data such as temp files, internet history, cookies and files in the systems recycling bin, this helps to free up space and keep the system’s

	 Below is a screenshot of a popup box of the Disk Cleanup clearing out the unwanted files of the computer system. 	performance from being decreased due to the excess of unused files. Deleting these unused and unnecessary files will help maintain a steady performance due to these files not slowing down the system. The second reason that this utility improves system performance is that it can also remove cookie files. Even though cookie files aren't a very significant problem to a computer system, but it can multiply at an alarming rate, especially if a user is using the internet frequently. Cookie files can prove to be dangerous when they aren't deleted constantly as that they can seriously impact a computer system's performance. To prevent this from happening, the cleanup tool that Windows has pre-installed will remove internet cookie files that are logged in a computer system, this should be done frequently due to the build up of cookies over time.
Disk Defragmentation	Disk Defragmenter Below is an image of the Disk Defragmenter showing the fragments of data within the disk drive. 	The first reason that Defragmentation improves system performance is that the utility frees up disk space and rearranges the fragmentations of data within a disk, which means that when trying to find a file the system can locate it easier than before. This increases system performance as it takes less time to find files and data. The second reason is due to the fact since defragmentation creates much more disk space which would improve the system by allowing it to run faster and also effectively extends the amount of data that can be stored on the storage media by rearranging the fragmentations and freeing up space.

<table>
<tr><td></td><td>Below is a screenshot of the Disk Defragmenter showing the disk drive after it has been properly defragmented and neatly set out, so the system can access the files much faster.
 
</td><td></td></tr>
<tr><td>Disk Management</td><td>Disk Management

Below is an image of a popup box showing if the user wants to format the F: drive of the system.
 

Below is a screenshot of the Disk Management showing all the disk drives within the pc, and they are all in healthy status.
 
</td><td>The first reason that disk management is good for improving the performance is that you can use it for drive formatting. What this is is basically to completely wipe out the data from a system, known as a factory reset. There are other methods of drive formatting which doesn't only consist of wiping the whole disk drive, you can have the option to only wipe one part of a hard drive, but it's advised to remove important information from the drive that is about to be formatted.

Another reason is that with this software you can remove certain programs and files that can have a massive impact on the performance of the computer system, such as software with large capacity or has a huge impact on boot up speed, which has a significant amount of impact to the performance of computer system.</td></tr>
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